

curriculum vitae

Nuclear Materials Modeling Specialist with 5+ years of experience analyzing reactor materials under extreme conditions (high temperature, neutron fluence). Expertise in finite element analysis, multi-scale computational mechanics, and verification/validation/uncertainty quantification for nuclear fuel performance applications. Combines deep materials science knowledge with computational modeling to deliver pragmatic engineering solutions for nuclear systems.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. MSc. BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed multiple surrogate models and custom Python-based tools to bridge polycrystalline mechanics of nuclear reactor materials (advanced alloys) to engineering-scale predictions under extreme conditions, such as high-temperature and irradiation.
- ❖ Developed multiple surrogate models and custom Python-based tools to bridge polycrystalline mechanics of nuclear reactor materials (HT-9, Grade 91 steel) to engineering-scale predictions under high-temperature and neutron fluence conditions.
- ❖ Developed multiple surrogate models and custom Python-based tools to bridge polycrystalline mechanics of nuclear reactor materials (HT-9, Grade 91 steel) to engineering-scale predictions under high-temperature and neutron fluence conditions.
- ❖ Collaborated with external industrial partners and national lab colleagues to deliver pragmatic modeling solutions; led technical milestone reports, peer-reviewed papers, and presented surrogate models at scientific conferences.
- ❖ Reached the R&D 100 Award Finals twice with the surrogate modeling solutions for nuclear materials applications.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

Developed the surrogate model ("LAROMance") for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low-length-scale simulations to enable engineering-scale structural predictions.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and validated advanced metrology systems for high-precision nanoscale defect detection, performing specialized cleanroom testing and data analysis under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

*Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.
Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands*

Designed and implemented an integrated computational-experimental framework combining finite element modeling, cohesive zone modeling of interfaces (mixed-mode), and inverse optimization techniques to characterize material system behavior, resulting in five journal publications.

2017 – 2020

Account manager & Technical recruiter

Research-Square & EuFlex | Eindhoven, The Netherlands

Acquired and managed client accounts, and conducted technical recruitment of highly specialized engineering and PhD-level candidates for science and technology companies.

2018 – 2019

Lab technologist, Multi-Scale Laboratory

Eindhoven University of Technology | Eindhoven, The Netherlands

Computational and Experimental Mechanics

Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.

2009 – 2013

Tutor of students

Eindhoven University of Technology, Mechanical Engineering Department | Freelance

Mentored bachelor student teams on engineering case studies, supporting balanced group dynamics, resolving team issues, and delivering constructive feedback.

Education

Eindhoven University of Technology, the Netherlands

Department of Mechanical Engineering

Division: Computational and Experimental Mechanics

Research Group: Mechanics of Materials

2013 – 2019

PhD.-research project

Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019

Graduate School of Engineering Mechanics

Advanced Mechanics Courses

2010 – 2013

MSc. Mechanical Engineering

Minor: Biomedical Engineering

2006 – 2010

BSc. Mechanical Engineering

2000 – 2006

Research-oriented secondary education (VWO)

St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. *Integrating Materials and Manufacturing Innovation* 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), <https://doi.org/10.1007/s11340-020-00583-8>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.